classified according to the types of environment that is affected. They are mainly air, water and soil pollution.

13.1.1 Classification of Pollutants

In terms of eco-system, pollutants can be classified into two basic groups – Non-degradable and degradable. Based on the time taken to breakdown into their ingredients, degradable pollutants are classified as rapidly degradable (non-persistent) and slowly degradable (persistent).

a) Rapidly degradable or non-persistent pollutants: These can be broken down by natural processes. Domestic sewage and vegetable waste are examples of such pollutants.

b) Slowly degradable or persistent pollutants: These are pollutants that remain in the environment for many years in an unchanged condition and take decades or longer to degrade, as in the case of DDT.

c) Non-degradable pollutants: These cannot be degraded by natural processes. Once they are released into the environment, they are difficult to be eliminated and continue to accumulate (biomagnification). Toxic elements like lead, mercury, cadmium, chromium and nickel are such common pollutants.

13.2 Air Pollution

Earth is surrounded by a gaseous envelope which is called atmosphere. The gaseous blanket of the atmosphere acts as a thermal insulator and regulates the temperature of the earth by selectively absorbing The UV rays of solar radiation. The adverse effects of pollution include depletion of Ozone by Chlorofluorocarbons or CFCs, used as refrigerants and global warming by elevated CO_2 (industries, deforestation, and partial combustion).

The alterations or changes in the composition of the earth's atmosphere by natural or human activities (anthropogenic factors) are referred as **Air Pollution**. Pollutants include the abundant presence of solid, liquid or gaseous substances produced by human or natural activity. The nature and concentration of a pollutant determines the severity of detrimental effects

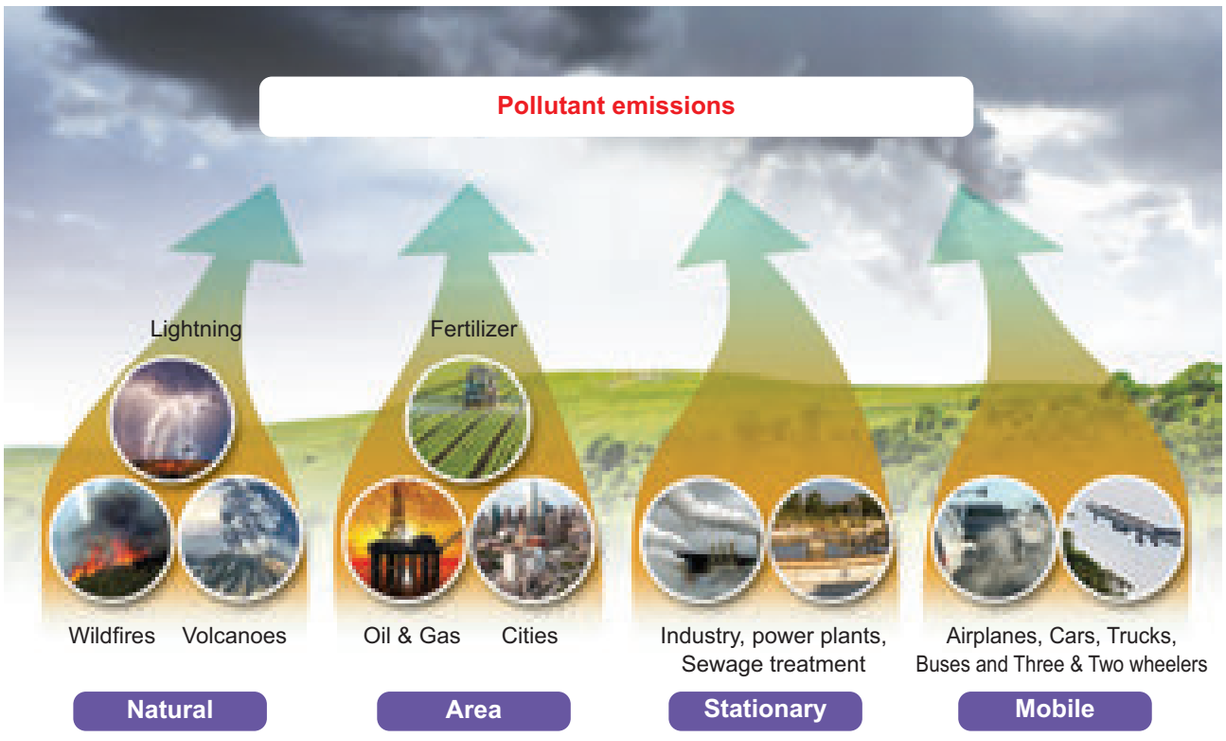


Fig. 13.1 Sources of air pollution.

TN_GOVT_XII_Zoology_chapter13.indd 220 24-02-2020 4.04.58 PM